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Principles of Biosignals and Biomedical Imaging

Project – 2025/26

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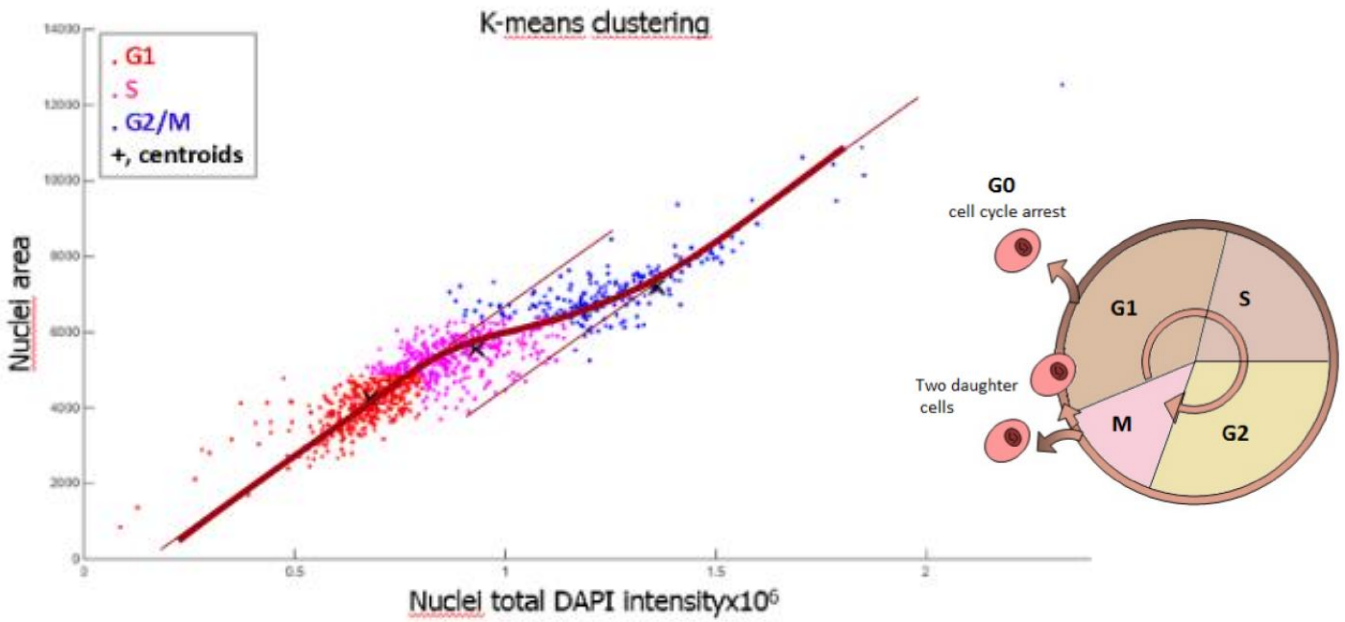
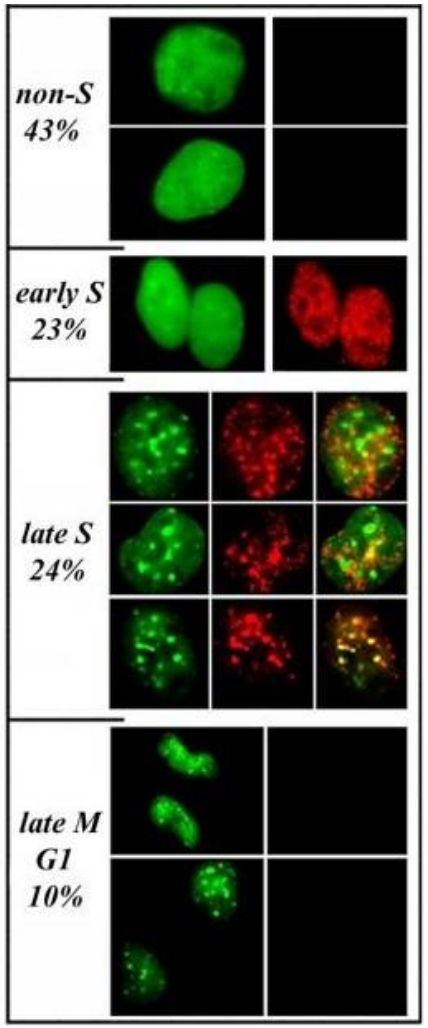
DAPI as a cell cycle profiler



Cell Cycle classification based on Nuclei Area



Differences within the nuclei throughout the cell cycle

Methodology

Objective: Classify cells into their cell cycle stages using images and Machine Learning

Pre-Processing

Improve image quality (e.g. noise, intensity, contrast)

Image Processing

Segment nuclei and obtain nuclei masks

Feature Extraction

Extract nuclear area, blue color intensity, texture features, etc.

Classifier

Use a model to classify the different nuclei into G1 or G2 phases

Based on Ferro, Anabela, et al. "Blue intensity matters for cell cycle profiling in fluorescence DAPI-stained images." *Laboratory Investigation* 97.5 (2017): 615-625.

Deliverables

- Deadline: 5th April 2026 (23.59 PM)
- 4-page report (e.g. IEEE Conference template)
- Project code
- Presentation (5 min.) + Questions (10 min.): 8th and 9th April 2026

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